

SurfaceLayerDiffusivity: GABLS1 results

Corrected transport comparison | 21 September 2026 | four matched runs, one seed.

Question: can transport confined to one or two near-wall faces improve coarse LES? Four paired GABLS1 runs use WENO9 at 12.5 m resolution (32^3 cells), a 400 m cube, the same frozen source, and 9 h of integration.

Finding: all three SurfaceLayerDiffusivity variants move the mean u , mean θ and w^2 profiles farther from the fixed 1 m LES median by RMS error over native levels below 200 m. This result applies to this grid and these settings; it is not a general verdict on the closure.

Near-wall response: at 12.5 m, final-hour w^2 falls by 85–93%. Skewness changes from +0.56 in the control to -0.21, -0.25 and -0.42. Domain-integrated resolved TKE falls by 15–20%. The first-face variance is suppressed much more strongly than the integrated energy.

Surface exchange: final-hour u^* rises from 0.279 m/s to 0.292, 0.289 and 0.301 m/s, versus a reference mean of 0.262 m/s. Surface cooling also strengthens, moving farther from the reference. One-face 100 and 300 s responses are similar; two-face support increases local suppression.

Smagorinsky comparison: the earlier 12.5 m WENO9 run ($C_s=0.16$, $Pr=1$) has a final-hour peak w^2 of only $1.12e-6 \text{ m}^2/\text{s}^2$, versus 0.0864 in the control. Its mean- u error is smaller despite almost absent resolved turbulence. Teal curves show this historical context; its source differs from the four matched runs. Log axes reveal the contrast.

Next hypothesis: restrict SLD viscosity to u and v , retaining tracer diffusivity and all other settings. This isolates direct damping of w , but indirect effects through shear and stratification remain. No u/v -only result is included. Single-seed comparisons do not establish uncertainty; the LES median is not observational truth.

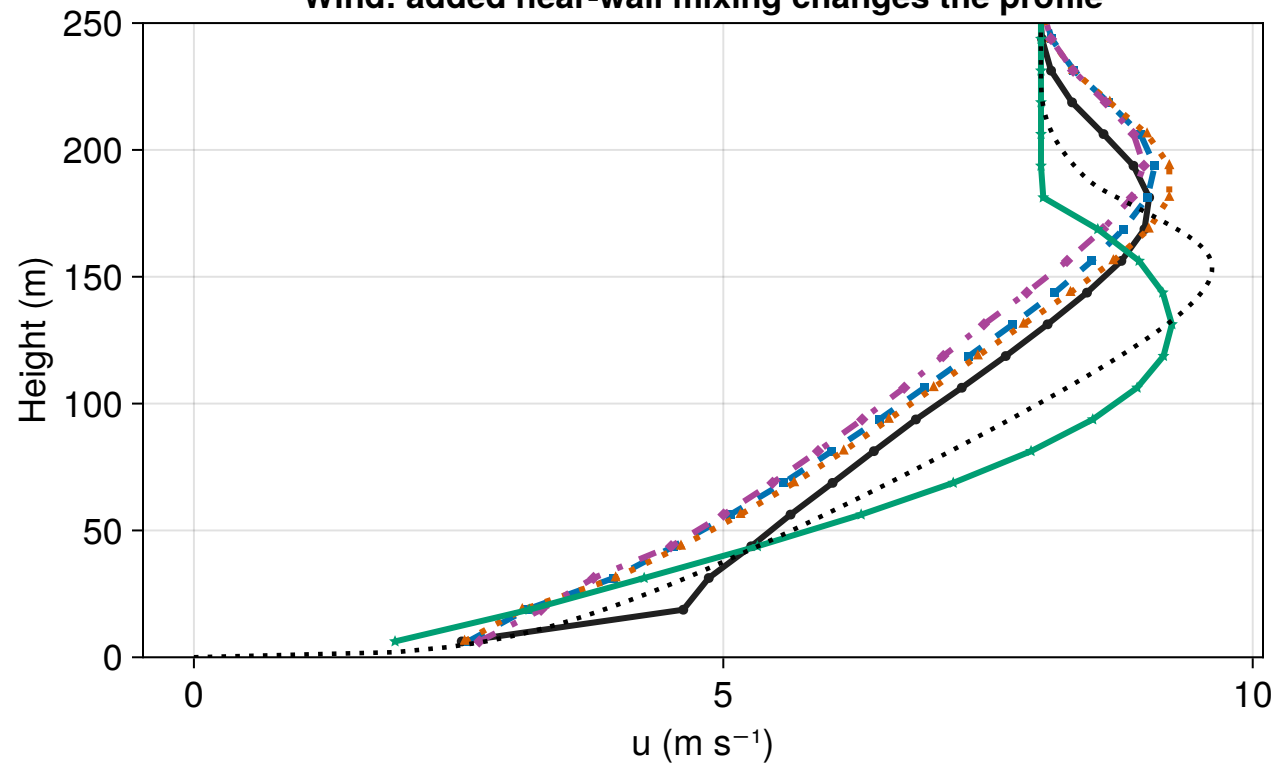
Transport split: corrected diagnostics show SGS mixing supplies 86–95% of first-level u -momentum transport in the SLD cases. Total flux magnitude rises by only 6–17%, while its resolved contribution falls sharply. Mean fields and resolved moments exactly reproduce the earlier runs. The new native coefficient–gradient products pass GPU and export consistency checks; this split does not measure WENO numerical mixing. Historical array 7156 fluxes remain excluded; current figures use corrected array 7293.

Reading the figures: native-height profiles are true 8–9 h means; timelines use instantaneous output. Colors, line styles and markers distinguish cases. Exact data, Julia source, reference hashes and diagnostic exclusions accompany this brief; all original DYCOMS/GABLS1 report material is retained.

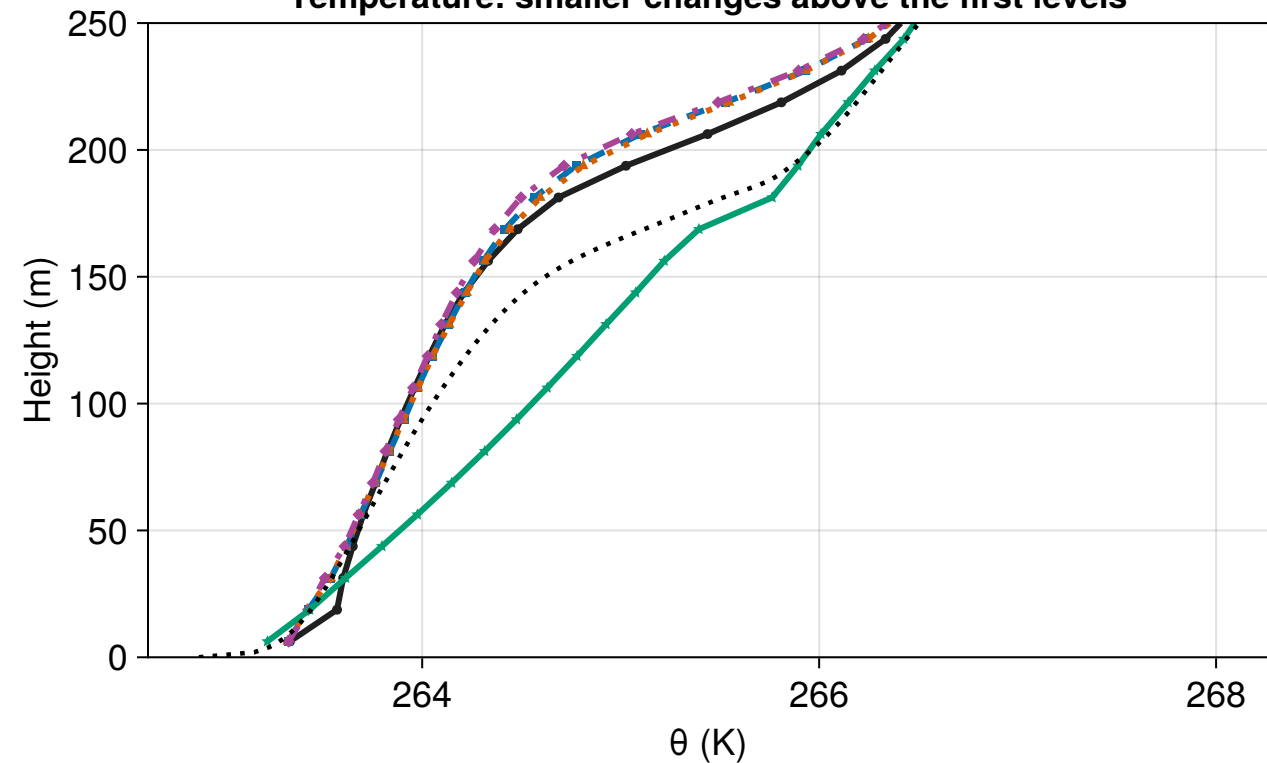
Surface-layer mixing versus coarse-grid Smagorinsky

GABLS1 · WENO9 · 12.5 m grid (32^3) · final hour, 8–9 h · four matched runs + earlier Smagorinsky

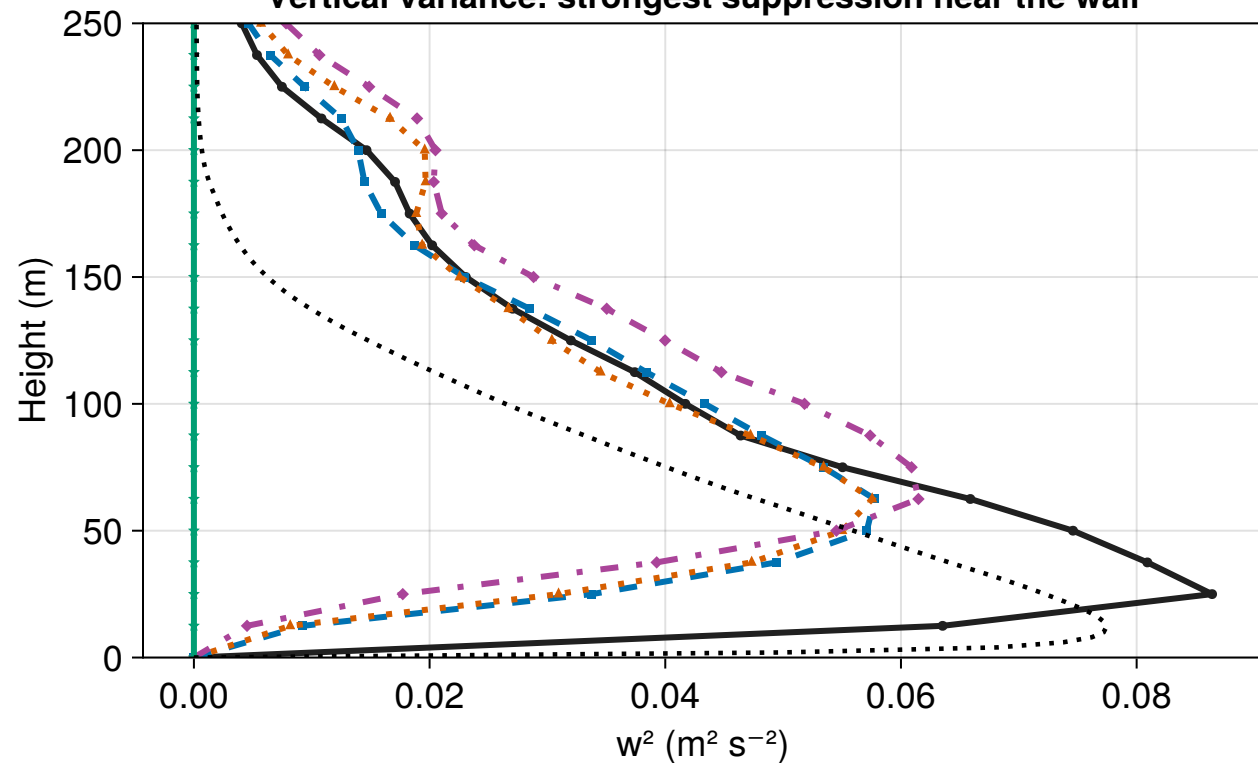
Wind: added near-wall mixing changes the profile



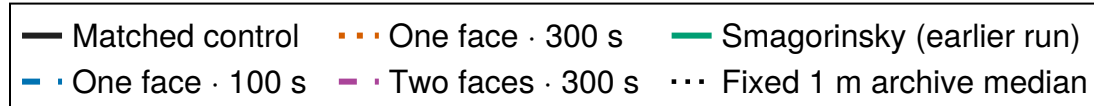
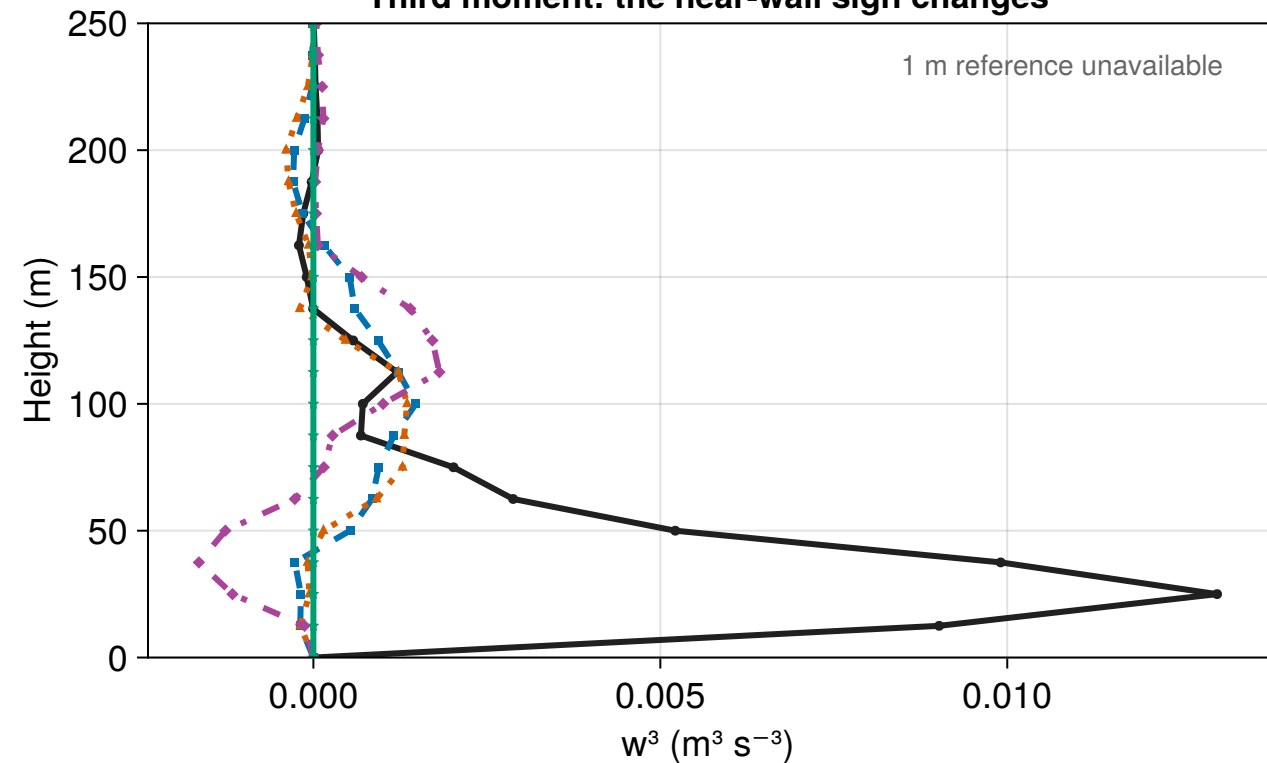
Temperature: smaller changes above the first levels



Vertical variance: strongest suppression near the wall



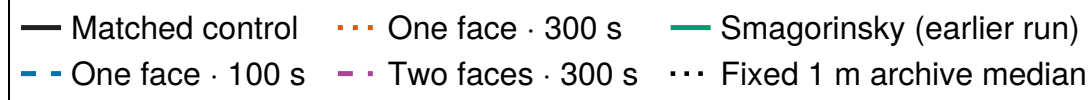
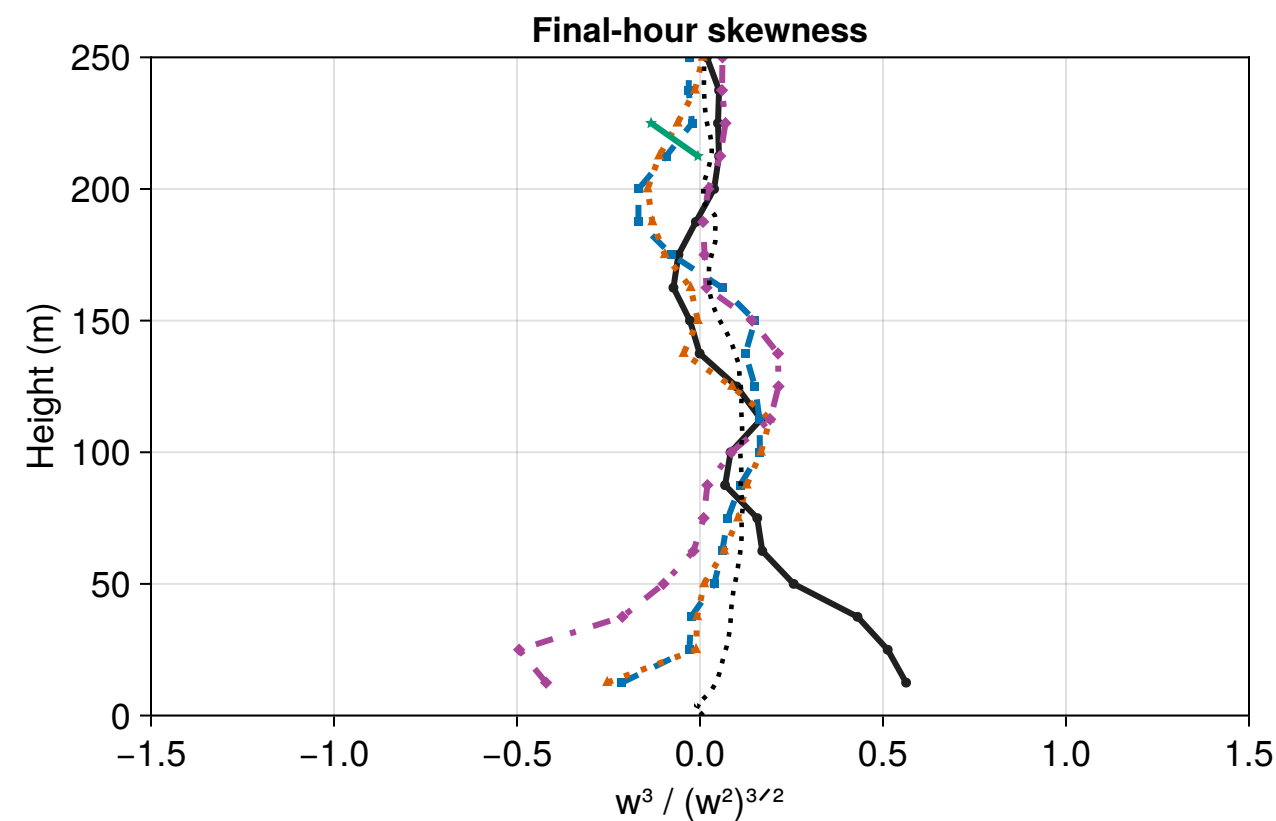
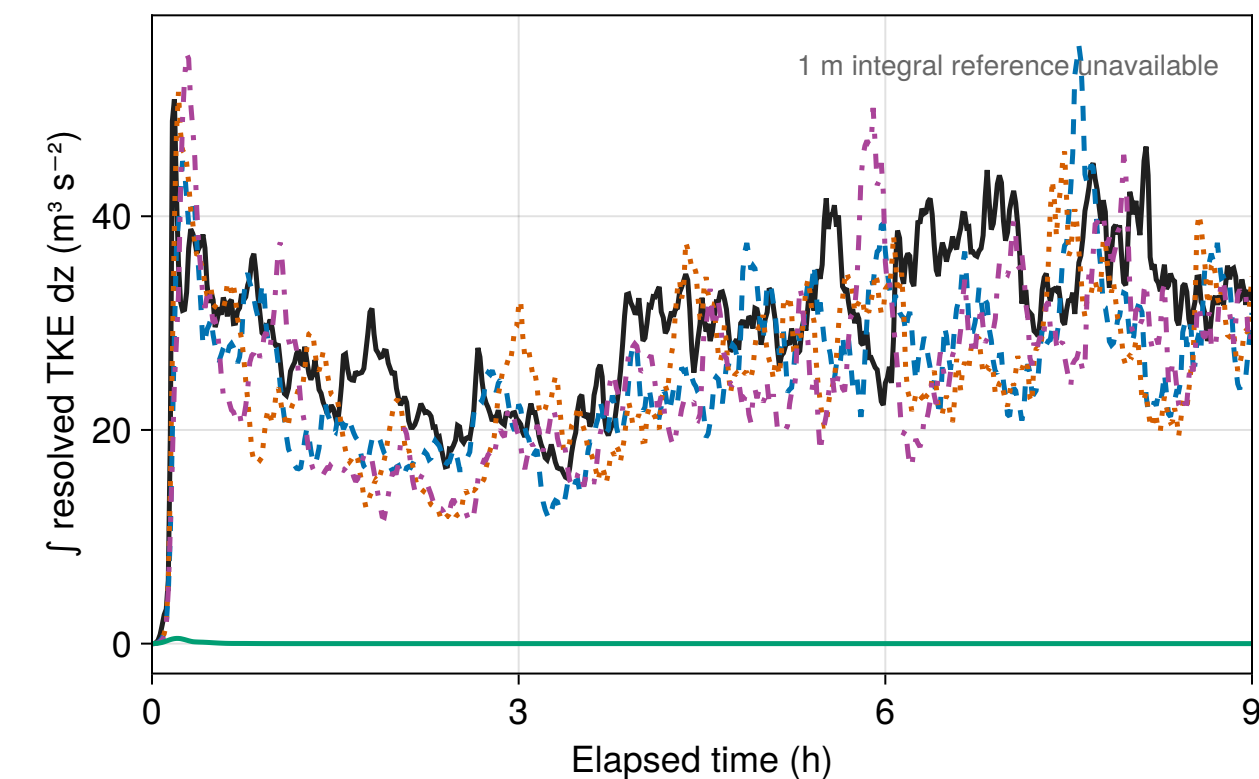
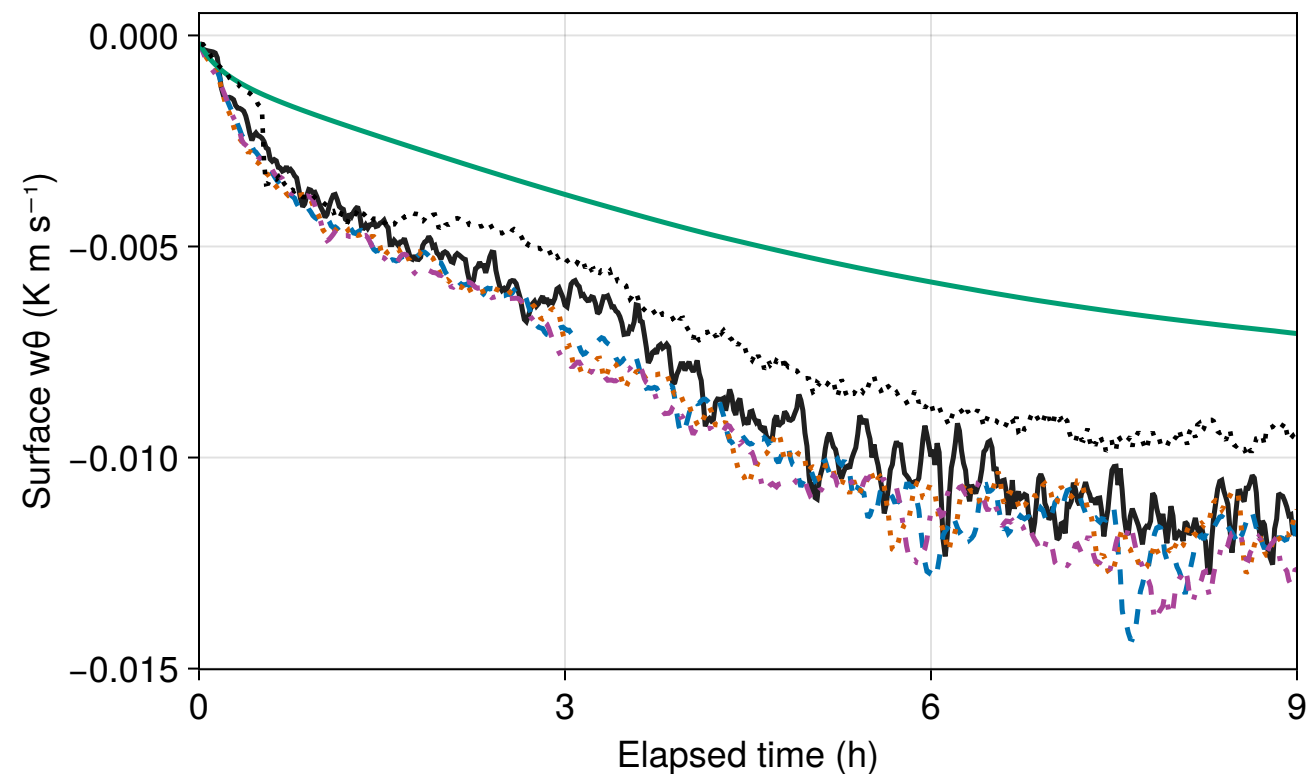
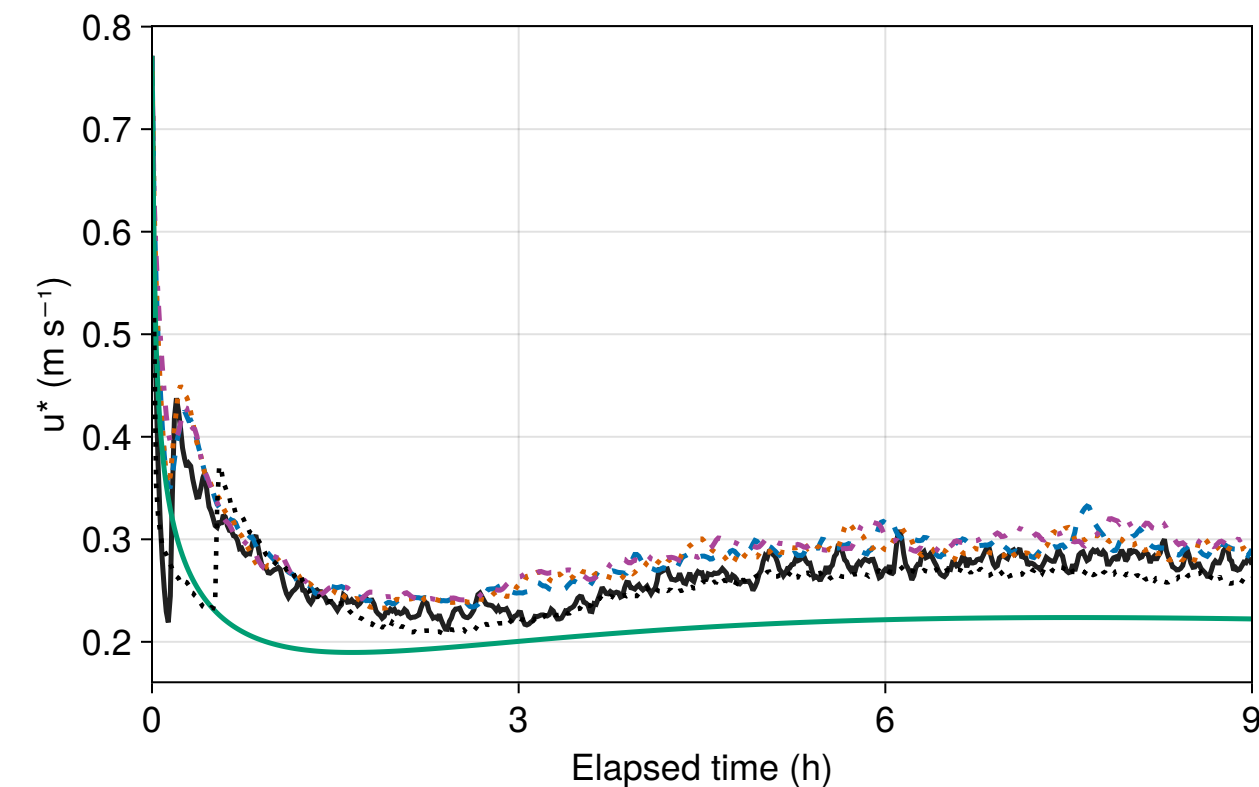
Third moment: the near-wall sign changes



Teal stars: earlier Smagorinsky source; its w^2 and w^3 nearly coincide with zero on these linear axes.
Black dotted: fixed 1 m LES median. Matched runs use the corrected, GPU-validated flux diagnostics.

Follow the response beyond the mean profile

Surface exchange through 9 h; skewness from the final-hour native-face moments

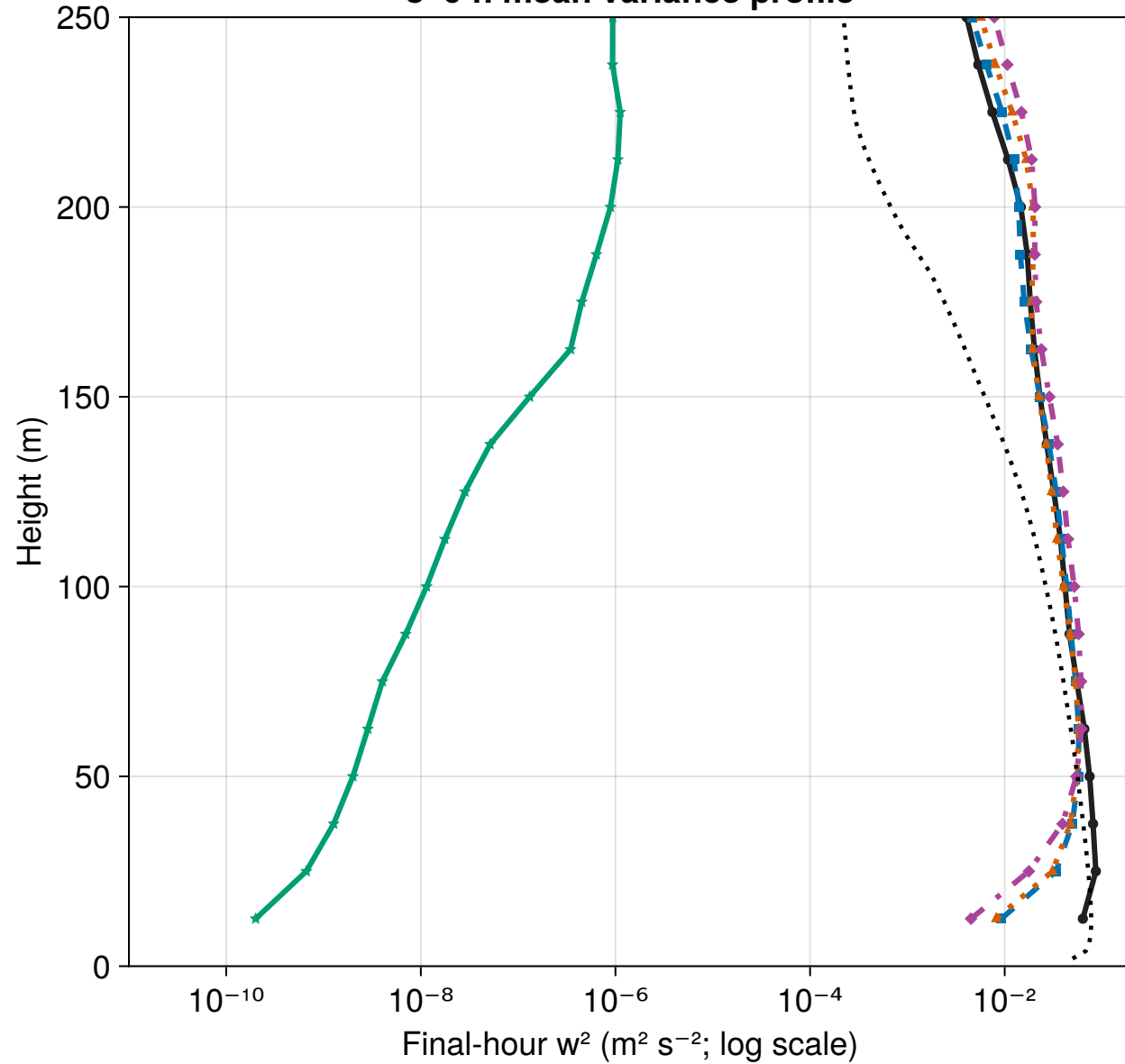


Wall flux remains usable; resolved TKE excludes SGS energy. Teal: earlier Smagorinsky source.
Skewness omits $w^2 < 10^{-6} \text{ m}^2 \text{s}^{-2}$, masking most Smagorinsky levels; a ratio of tiny moments is not robust.

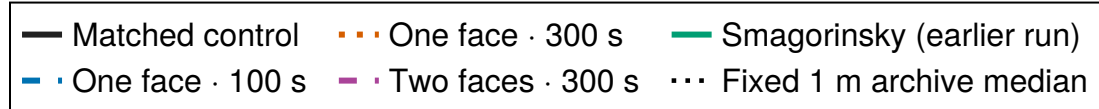
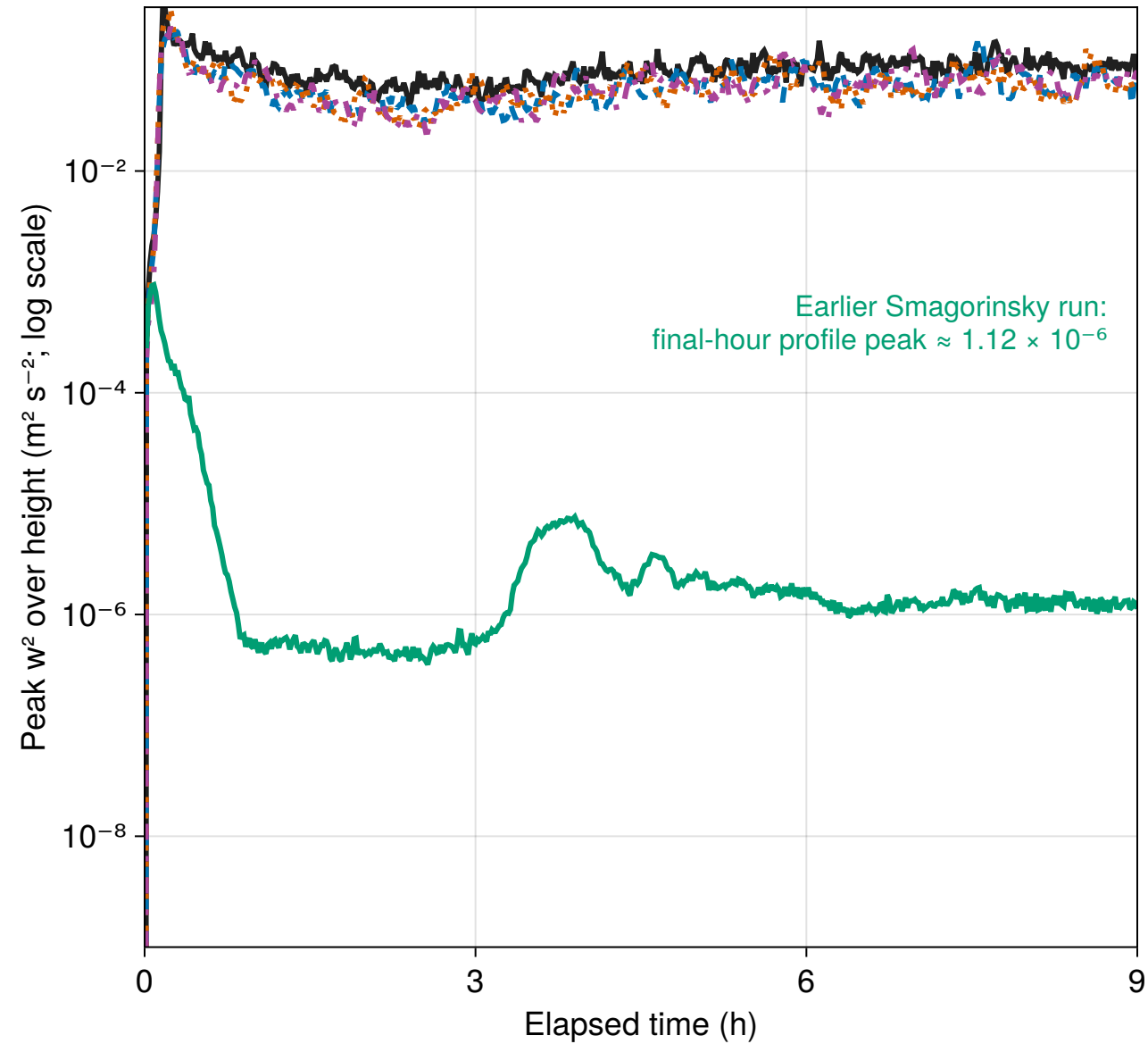
Smagorinsky nearly eliminates resolved vertical turbulence

12.5 m GABLS1 · WENO9 · log axes expose the difference hidden by linear scales

8–9 h mean variance profile



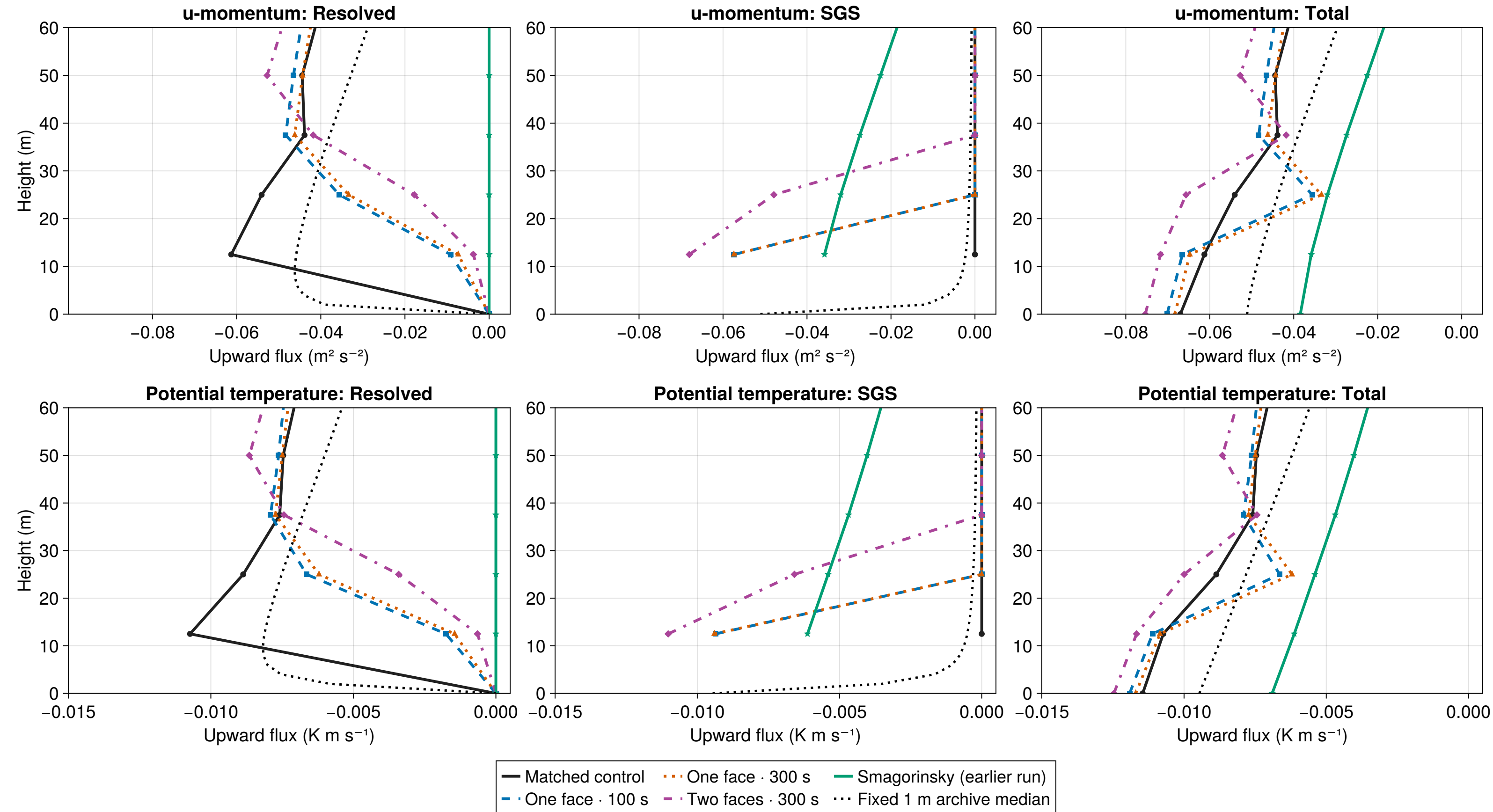
Instantaneous horizontal variance



Same grid and advection; Smagorinsky is historical context, not a same-revision controlled treatment.
Zeros are omitted on log axes. The left panel averages first; the right panel takes each instantaneous maximum.

SGS transport replaces most of the resolved near-wall flux

GABLS1 · 12.5 m · 8–9 h means · native faces in the lowest 60 m · corrected diagnostics



Colors, line styles and markers identify cases. SGS wall zeros are omitted; total uses the prescribed wall flux. Interior total = resolved + SGS. This split does not quantify WENO numerical mixing. Teal is historical context.

Resolved-flux factor 2: limited response

Matched sensitivity | 21 September 2026 | GABLS1, 12.5 m, WENO9, one interior face, 300 s filter, 9 h, one seed.

Result: crediting twice the resolved transport reduces the added diffusivity, but does not restore resolved turbulence. In the final hour, first-level w^2 rises 7.1%, while peak w^2 falls 6.3% and vertically integrated resolved TKE falls 6.4%. Both factors remain far below the earlier no-closure near-wall variance.

The intervention: factor a multiplies only the signed local filtered resolved flux inside the closure deficit $\max(0, 1 - a F_{\text{resolved}}/F_{\text{target}})$. The same factor is used for momentum and heat. Factor 2 assumes an additional numerical flux equal and aligned with resolved transport. It neither doubles viscosity nor rescales diagnostic fluxes. The experiment does not measure numerical transport or establish that assumption.

Mechanism: final-hour first-face viscosity falls from 1.321 to 1.118 m^2/s (15.4%); heat diffusivity falls from 1.216 to 0.774 m^2/s (36.3%). The first-face SGS share of u-momentum transport remains large: 88.7% versus 86.8%. Reduced closure coefficients therefore produce only a modest change in the resolved/SGS partition.

Moments: at 12.5 m, w^2 changes from 0.008171 to 0.008754 m^2/s^2 , versus 0.06353 without interior closure. The third moment changes from $-1.872\text{e-}4$ to $-1.186\text{e-}4$ m^3/s^3 , and skewness from -0.253 to -0.145. Skewness is the ratio of hourly averaged moments, not an average of instantaneous skewness.

Mean profiles and exchange: u^* falls from 0.2886 to 0.2831 m/s and sensible heat flux changes from -15.54 to -15.38 W/m^2 . Against the fixed 1 m median at native heights $0 < z \leq 200$ m, final-hour RMS errors change by +0.3% for u , -9.2% for θ , and -7.3% for w^2 . Smaller profile errors coexist with lower resolved energy; they do not demonstrate restored turbulence.

Hour sensitivity matters: in 7–8 h, first-level w^2 instead falls 5.1%, integrated TKE falls 10.4%, and the temperature-profile error increases 8.0%. Peak variance and integrated energy decrease in both windows. The near-wall variance increase and temperature improvement in the last hour are not consistent across the two hours. One seed and neighboring hours do not provide an uncertainty estimate.

Comparison integrity: two fresh cases passed strict admission with matching source, grid, seed, initialization and exact 19-profile/541-series schedules. The factor-1 run reproduces every exported profile and time-series value from corrected array 7293's one-face300 case bit for bit. Native physical flux partition, support and coefficients were checked. Fixed 1 m medians are LES references, not observations.

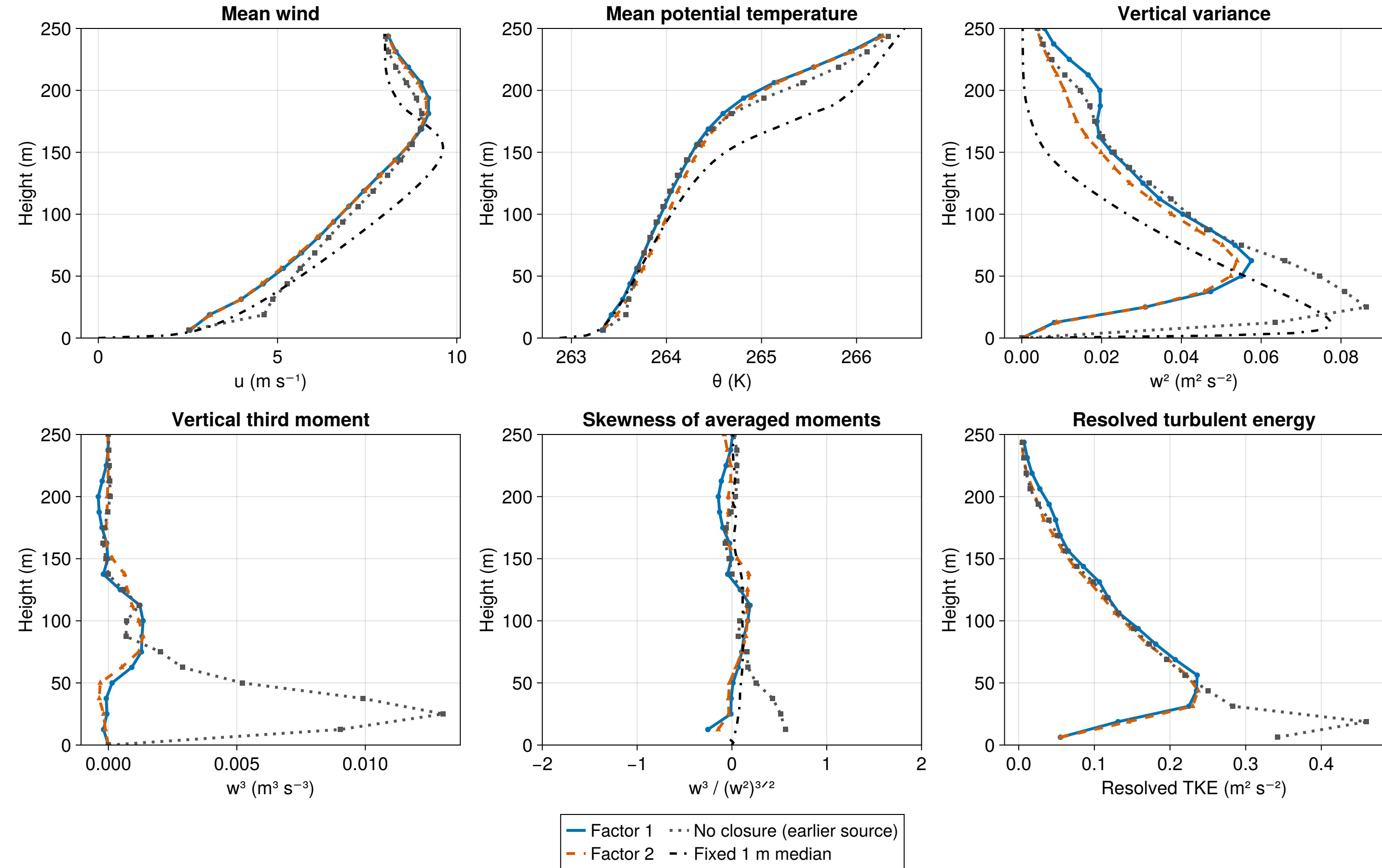
Resolved-flux factor 2: limited response

Execution provenance: both model children completed 32400 s and exited zero. Slurm nevertheless marked the factor-2 batch exit as one. A separate harmless login-shell probe reproduces that pattern through failing `clear_console` in `.bash_logout` under `set-e`; actual batch shell depth was not recorded. The discrepancy and canceled downstream job are preserved. Admission relies on independently complete outputs, durable child exits, matching sources and logs, not a relabeled scheduler success.

Conclusion: this factor-2 test changes coefficients and exchange modestly, but provides no evidence of broad turbulence recovery. It does not identify the numerical flux or establish a calibrated correction factor. All earlier DYCOMS, GABLS1 and SurfaceLayerDiffusivity results remain in the master report. GABLS3's separate humidity-boundary validation is still pending; no new GABLS3 science is included.

Crediting twice the resolved flux: does turbulence recover?

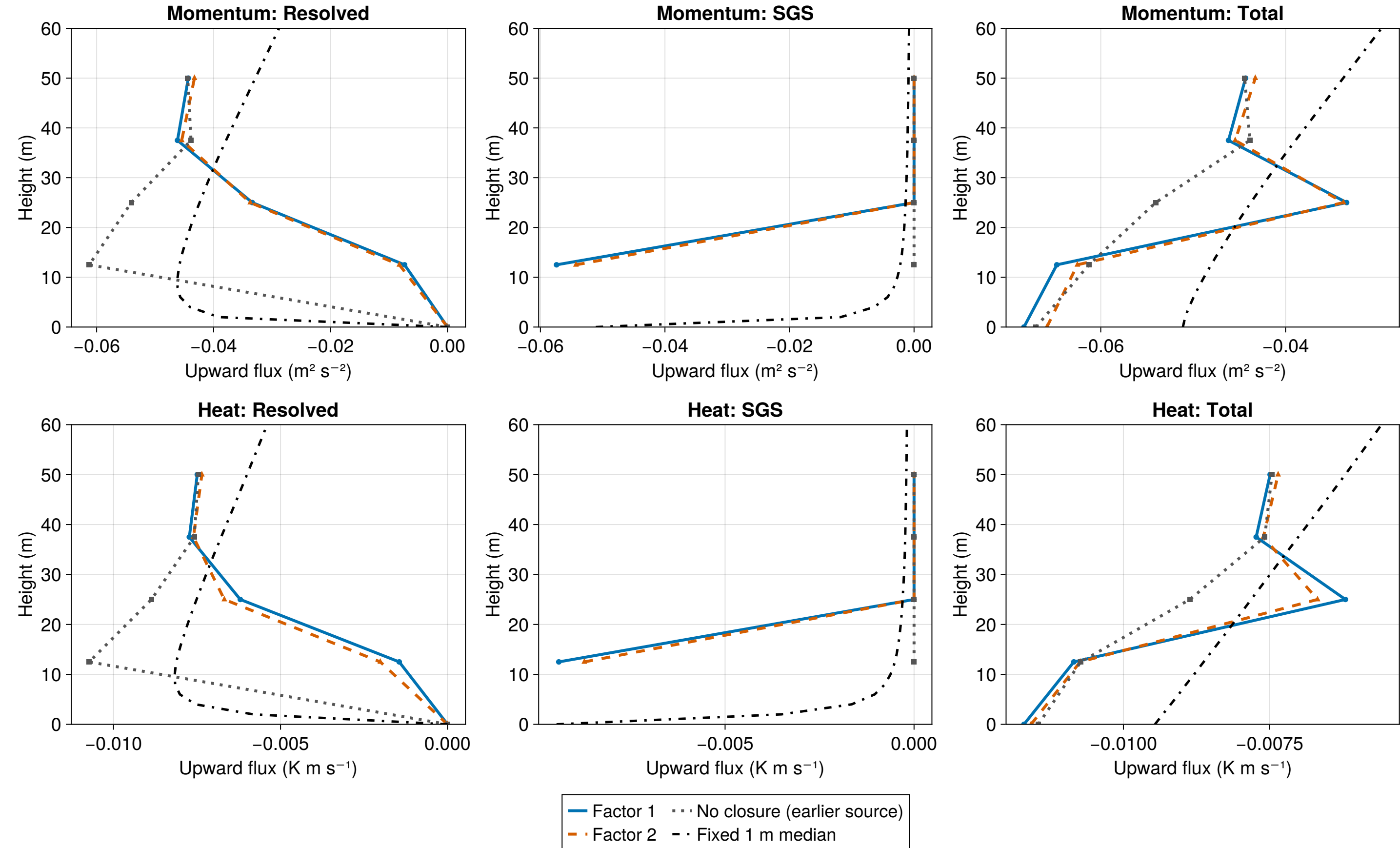
GABLS1 · 12.5 m · WENO9 · one face · 300 s filter · 8–9 h mean



Factor changes the closure deficit for momentum and heat; displayed physical fluxes are unscaled.
Earlier no-closure run is context. This tests an assumed numerical contribution, not a measured numerical flux.

Resolved and SGS transport near the surface

Same final-hour average · native flux faces · physical fluxes remain unscaled



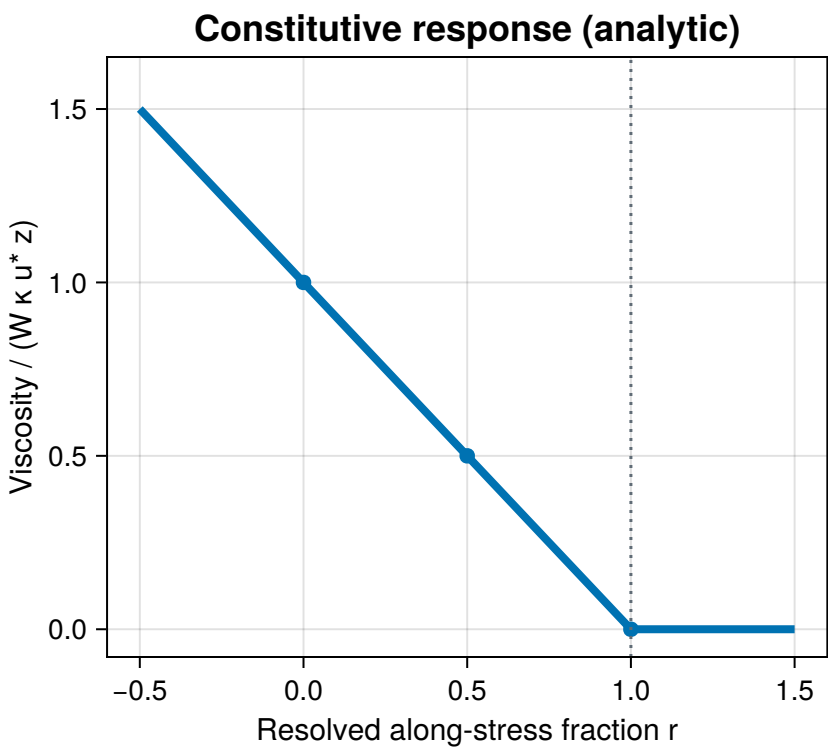
Interior total = resolved + SGS. At the wall, total includes the prescribed flux; artificial SGS wall zeros are omitted.
Each panel includes the fixed 1 m median wherever that reference quantity is available.

Can a thin surface closure improve coarse LES?

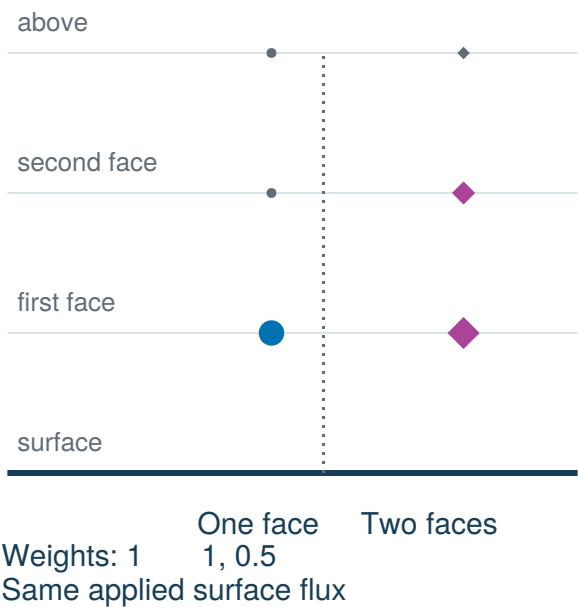
Supply unresolved near-wall transport while preserving resolved turbulence. Four coarse GABLS1 comparisons admitted; transfer tests ongoing.

A local estimate of missing transport

Filter velocity, scalar, and wall-flux samples in physical time. Use their centered covariance to estimate the resolved share of the surface flux.



Shallow vertical support



What this formula does

No resolved stress: full neutral similarity viscosity. Half resolved: half the viscosity. Fully resolved or more: the correction turns off. Opposite-sign transport can give a factor greater than one.

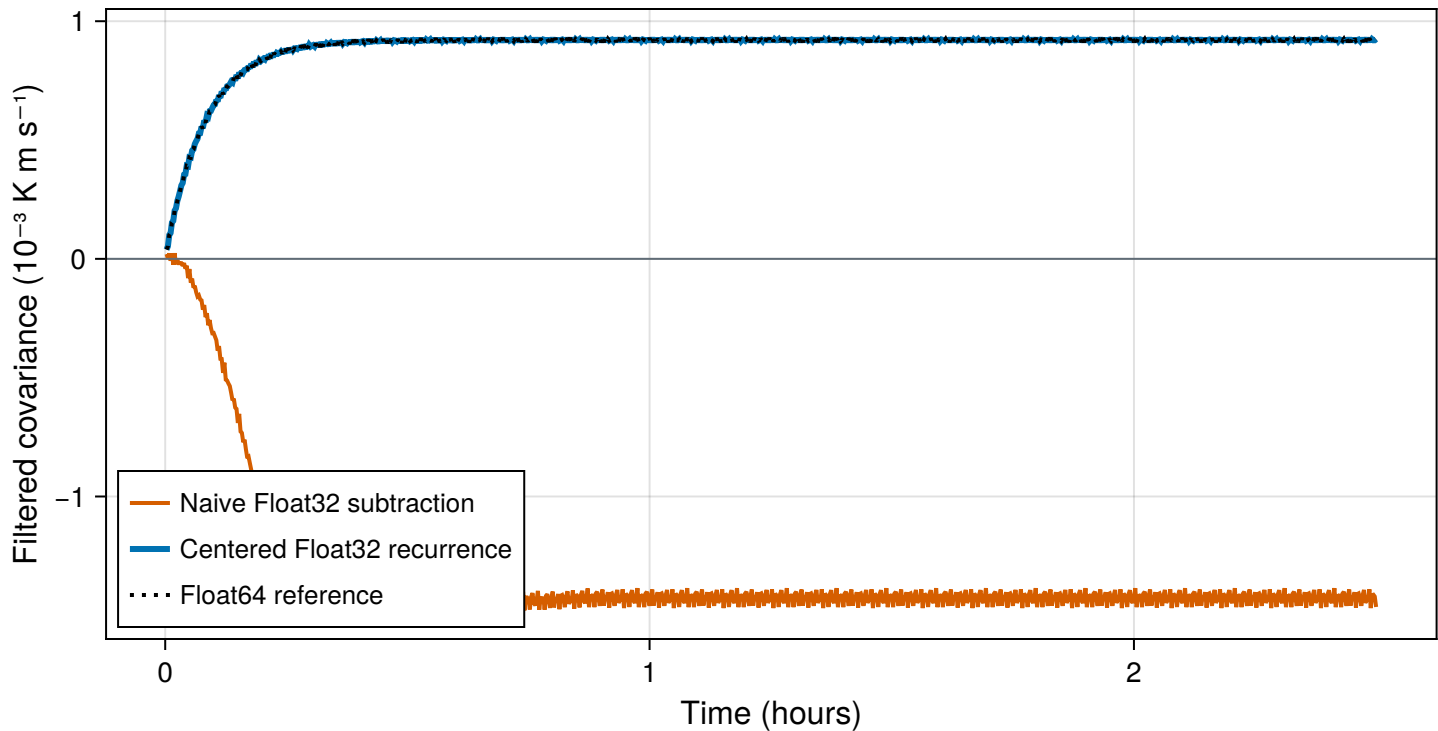
What it does not establish

The added flux still depends on the instantaneous vertical gradient. The formula does not force exact flux balance, include stability functions, or measure the numerical transport produced by WENO.

One-face height: 12.5 m in both planned GABLS comparisons. Scalar correction uses signed flux ratios; zero/tiny wall flux disables it.

A small covariance can get the wrong sign in Float32

Synthetic arithmetic test: 90,000 samples, $\Delta t = 0.1$ s, filter time = 300 s.
Identical Float32 input samples are used in the Float64 reference.



Final covariance: naive -1.465 | centered 0.925 | reference 0.924

Units: $10^{-3} \text{ K m s}^{-1}$. Centered recurrence relative error: 0.020%.

Why this matters for the closure

Subtracting two much larger filtered products loses the small turbulent flux. Here it reverses its sign. A centered covariance update retains the signal that controls the added diffusivity.

Implementation checks support the arithmetic result

The original 119-check CPU suite covers filtering, signs, guards and conservation. Later halo/restart checks and the 1,678-check GPU suite support the admitted run source. Their evidence files retain each revision.

Boundary of the evidence: this arithmetic example is synthetic.
Physical fidelity is assessed separately using the completed GABLS1 runs.
Reproducer, exact plotted CSV, and source hashes accompany this section.

Judge transport, turbulence, and the mean state together

One controlled intervention, three complementary tests of physical behavior. See the accompanying GABLS1 results for the first completed comparison.

01 Did the near-wall transport change as intended?

Lead with resolved, added, and combined momentum/heat flux profiles. Zoom the lowest four levels; mark every active face. Pair flux mismatch with viscosity, diffusivity, guard activity, and mean vertical transport.

02 Did the boundary layer improve without losing turbulence?

Place mean wind and temperature beside w^2 , w^3 and skewness. Show surface flux through time; withhold invalid depth diagnostics. Retain the fixed 1 m GABLS1 median; missing reference quantities remain explicit.

03 Is the result robust to filter time and support?

Use the same four styles in every GABLS panel. Compare with a freshly run, same-revision control, then examine 7-8 h versus 8-9 h. Report changes in physical units; avoid a composite score that hides tradeoffs.



Independent challenges strengthen the conclusion

GABLS3: moist nocturnal evolution and the sunrise flux sign change.
Neutral ABL: fixed-stress transport, without a surface-drag prediction.
Both add evidence; neither substitutes for the GABLS1 reference comparison.